

EE 446: TinyML

Lab 4: Visual Wake Words

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ENGINEERING**

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Lab #4

- Both Software and Hardware Lab!
- Develop a MobileNet model for Mask-Detection using **Edge Impulse**
- Using Transfer Learning to Train a Mask-Detection Model
- Compress MobileNet model for TinyML
- Deploy mask detection model in the Arduino board



Lab 4 – Introduction to Edge Impulse

Edge Impulse

- No Coding Required
- Edge Computing
- Support Various Sensors

The screenshot shows the Edge Impulse website homepage. At the top, a blue banner reads "Edge Impulse Optimizes NVIDIA AI for the Edge. [Read the Announcement!](#)". Below this is a navigation bar with the "EDGE IMPULSE" logo, and links for "Product", "Solutions", "Developers", "Pricing", "Company", "Blog", "Login", and a "Get started" button. The main content area features the headline "Build. Train. Optimize. AI for the edge." followed by a sub-headline: "Build datasets, train models, and optimize libraries to run on any edge device, from extremely low-power MCUs to efficient Linux CPU targets and GPUs." Below the text are two buttons: "Get Started" and "Schedule a demo". To the right of the text are three visual elements: a smartwatch displaying a "Heart Rate" graph, a "Sleep analysis" chart showing a 7h 32min duration, and a "Vibration" graph with a "Motion anomaly" alert. A large image of a robotic arm is also visible on the right side of the page.

Lab 4 [Software] – Edge Impulse Log In

1. Go to Website: <https://edgeimpulse.com/> ; Then Log In and create a new project

Log in



Username or e-mail

1

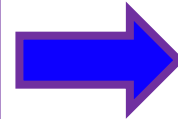


Password

[Forgot your password?](#)

Log in

Don't have an account? [Sign up](#)



+ Create new project

Lab 4 [Software] – New Project

2. Create a New Project

EDGE IMPULSE

DS

Dinuka Sahabandu

DEVELOPER

Enable MFA

Multi-factor authentication is now available for all users. [Set up now.](#)

Organizations

Allow anyone on your team to collaborate on multiple datasets, automation, and models in a shared workspace.

[Try Enterprise free](#)

Create a new project

Enter the name for your new project:

Mask Detection

Choose your project type:

- ☒ **Personal**
60 min job limit, 4GB or 4 hours of data, limited collaboration.
- ☐ **Enterprise**
No job or data size limits, higher performance, custom blocks.
- ☐ **Public**
Anyone on the internet can view and clone this project under the [3-Clause BSD license](#). Only invited users will be able to edit.

Choose your project setting:

- ☐ **Public**
- ☒ **Private (1 of 3 remaining)**
Only invited users can edit and view your project.

[Want full-feature access and unlimited projects? Try Enterprise free.](#)

Create new project

+ Create new project

Lab 4 [Software] – Choose Target Device

3. Scroll down and choose the target device first!

Dashboard

Devices

Data acquisition

Experiments

EON Tuner

Impulse design

- Create impulse
- Retrain model
- Live classification
- Model testing
- Deployment

Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

Project info

Mask

This is your Edge

+ New tag

Getting started

Start building

Add

Start with a

Not sure when

Target device

Define your target device requirements to inform model optimizations and performance calculations. No device yet? Use the default settings which you can change at any time.

Target device

Arduino Nano 33 BLE Sense (Cortex-M4F 64MHz)

Processor family

Cortex-M

Clock rate

64

Max

Custom device name (optional)

Application budget

Specify the available RAM and ROM for the model's operation, along with the maximum allowed latency for your specific application. Not sure yet? Start with the defaults and modify them later on.

RAM

256

Max

ROM

1

Max

Latency

100

Max

Reset to default settings

Cancel

Save

Target: Cortex-M4F 80MHz

DS

Private

This project is private, only invited users can edit and view.

Published versions (0)

This project has no published versions.

Publish a version of your project

W

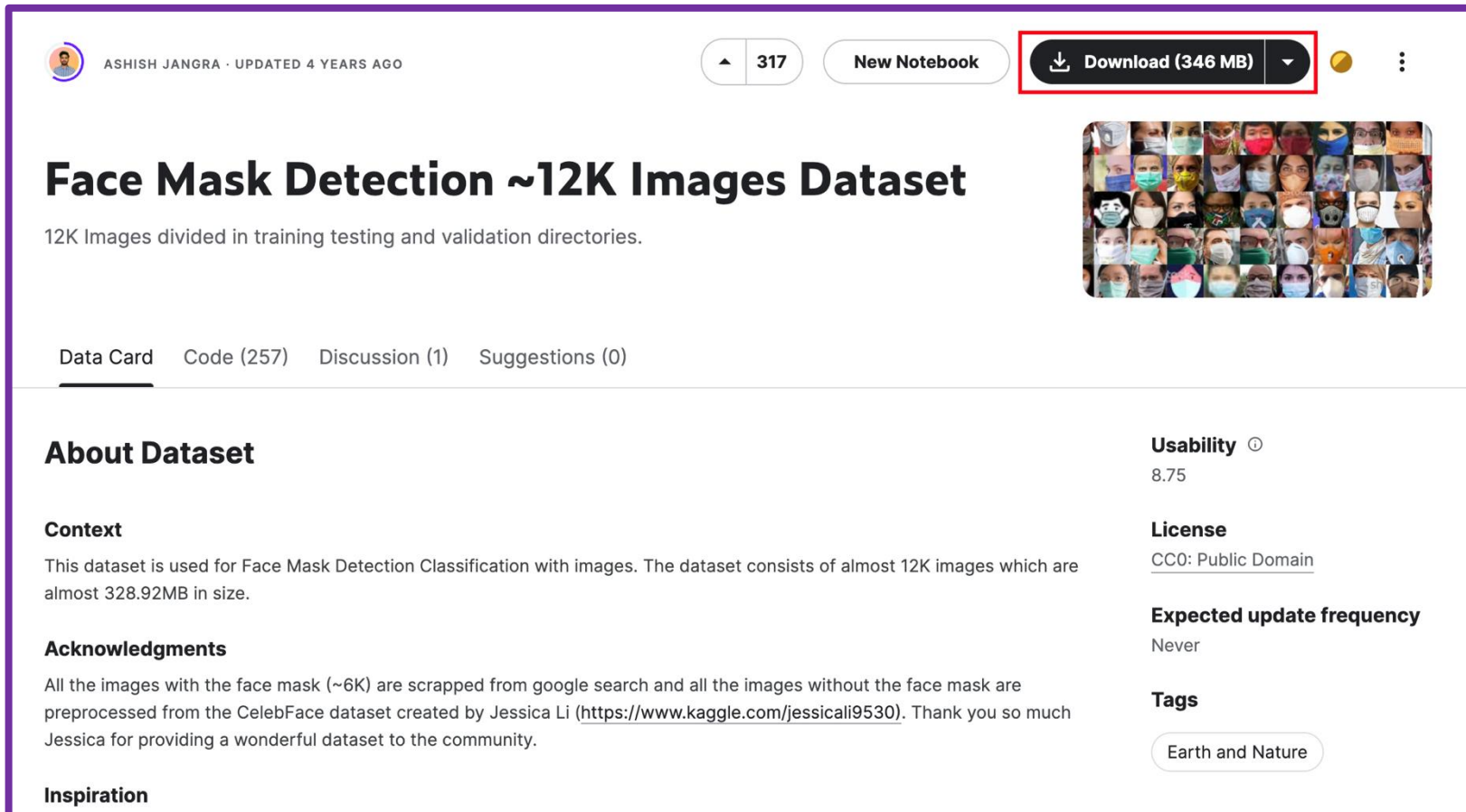
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Lab 4 [Software] – Download Dataset

4. Download the dataset: We obtain a publicly available dataset from Kaggle
<https://www.kaggle.com/datasets/ashishjangra27/face-mask-12k-images-dataset>



ASHISH JANGRA · UPDATED 4 YEARS AGO

317 New Notebook

Download (346 MB)

Face Mask Detection ~12K Images Dataset

12K Images divided in training testing and validation directories.

Data Card Code (257) Discussion (1) Suggestions (0)

About Dataset

Context

This dataset is used for Face Mask Detection Classification with images. The dataset consists of almost 12K images which are almost 328.92MB in size.

Acknowledgments

All the images with the face mask (~6K) are scrapped from google search and all the images without the face mask are preprocessed from the CelebFace dataset created by Jessica Li (<https://www.kaggle.com/jessicali9530>). Thank you so much Jessica for providing a wonderful dataset to the community.

Inspiration

Usability

8.75

License

CC0: Public Domain

Expected update frequency

Never

Tags

Earth and Nature

Lab 4 [Software] – Upload Dataset

5. Upload the dataset into your Edge Impulse project

In your Edge Impulse, click: “Add existing data”

The screenshot shows the Edge Impulse web interface for a project named "Mask Detection". The interface is divided into several sections:

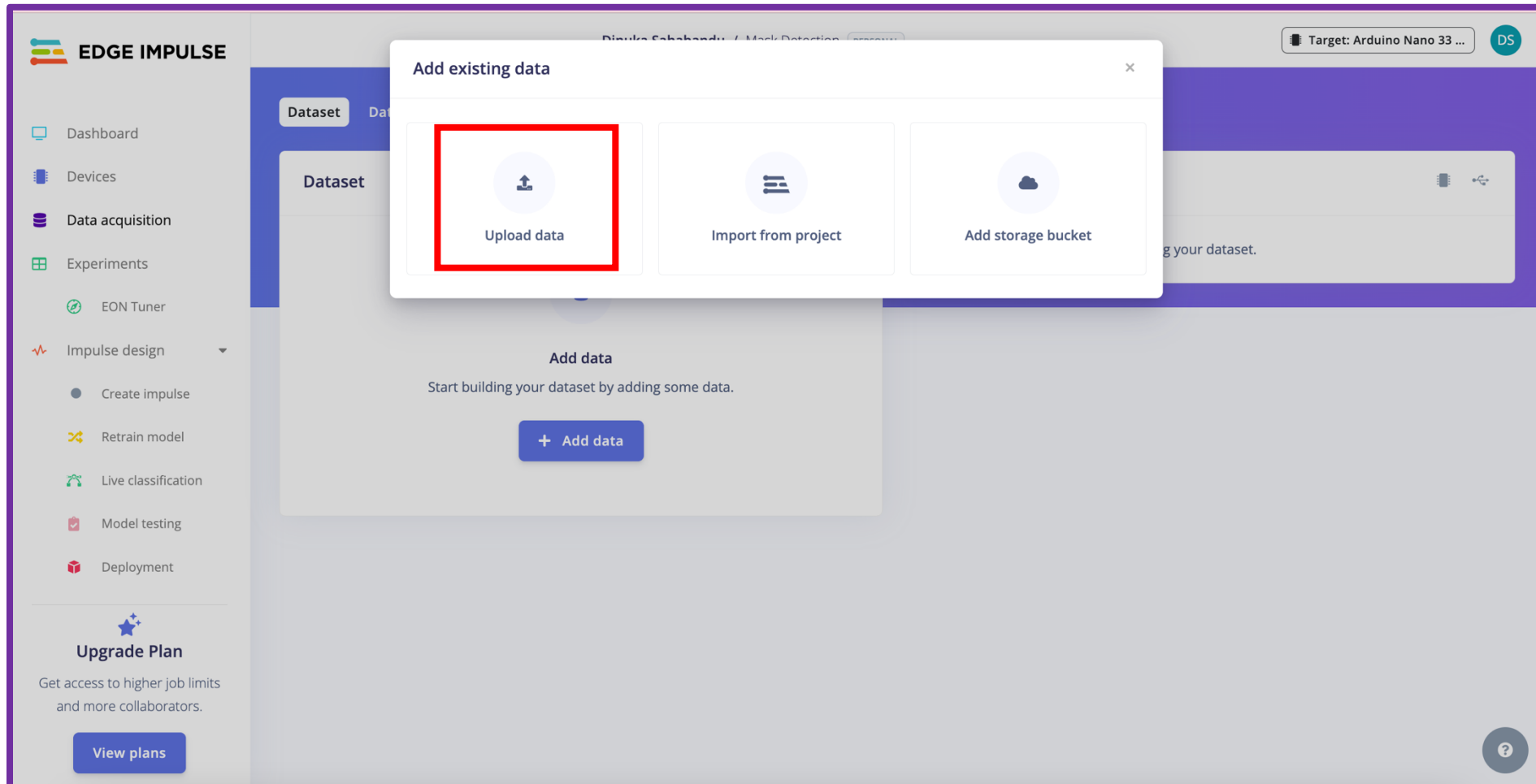
- Left Sidebar:** Contains navigation links for Dashboard, Devices, Data acquisition, Experiments, EON Tuner, and Impulse design. Under Impulse design, there are links for Create impulse, Retrain model, Live classification, Model testing, and Deployment. At the bottom, there is an "Upgrade Plan" section with a "View plans" button.
- Top Bar:** Displays the user name "Dinuka Sahabandu", the project name "Mask Detection", a "PERSONAL" tag, and a target device "Target: Arduino Nano 33 ...".
- Main Content Area:**
 - Project info:** The main header for the project, with tabs for Project info, Keys, Export, and Jobs.
 - Mask Detection:** The title of the project, followed by a description: "This is your Edge Impulse project. From here you acquire new training data, design impulses and train models." and a "+ New tag" button.
 - Getting started:** A section with the text "Start building your dataset or validate your model's on-device performance:". It contains three cards: "Add existing data" (highlighted with a red box), "Collect new data", and "Upload your model".
 - Start with a tutorial:** A section with the text "Not sure where to start? Follow a tutorial to build your first model in just minutes!".
 - Sharing:** A section with a "Private" dropdown menu and the text "This project is private, only invited users can edit and view."
 - Published versions (0):** A section with the text "This project has no published versions." and a "Publish a version of your project" button.

The URL at the bottom of the browser window is <https://studio.edgeimpulse.com/studio/739231/acquisition/training>.

Lab 4 [Software] – Upload Dataset

6. Upload the dataset into your Edge Impulse project

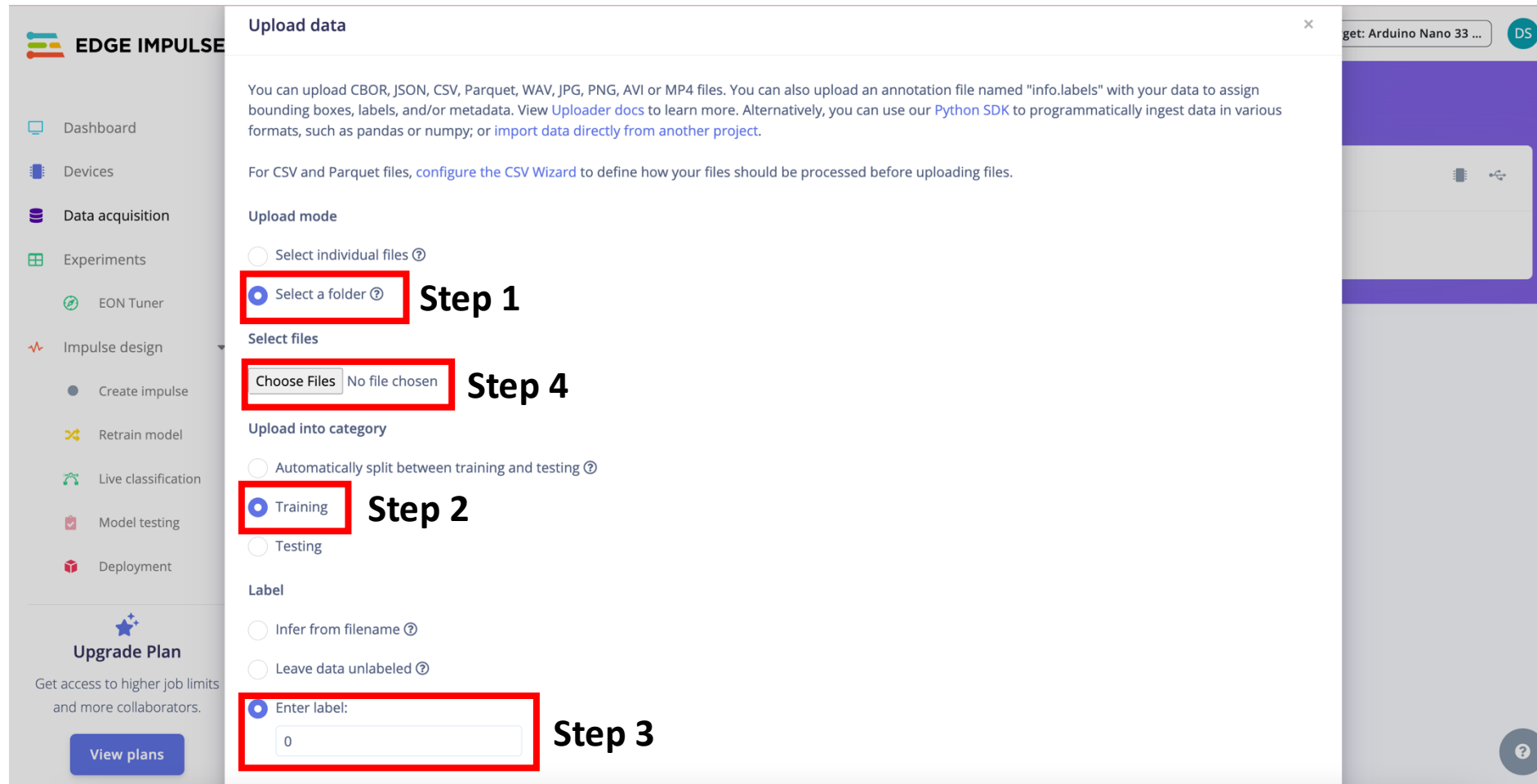
In your Edge Impulse, click: “Upload data”



Lab 4 [Software] – Upload Training Data Without Mask

7. Upload the “Training Data Without Mask” into your Edge Impulse project

Enter Label: “0”



EDGE IMPULSE

Dashboard
Devices
Data acquisition
Experiments
EON Tuner
Impulse design
Create impulse
Retrain model
Live classification
Model testing
Deployment

Upgrade Plan
Get access to higher job limits and more collaborators.
[View plans](#)

Upload data

You can upload CBOR, JSON, CSV, Parquet, WAV, JPG, PNG, AVI or MP4 files. You can also upload an annotation file named "info.labels" with your data to assign bounding boxes, labels, and/or metadata. View [Uploader docs](#) to learn more. Alternatively, you can use our [Python SDK](#) to programmatically ingest data in various formats, such as pandas or numpy; or [import data directly from another project](#).

For CSV and Parquet files, [configure the CSV Wizard](#) to define how your files should be processed before uploading files.

Upload mode

☐ Select individual files ?

☒ **Select a folder ?** **Step 1**

Select files

No file chosen **Step 4**

Upload into category

☐ Automatically split between training and testing ?

☒ **Training** **Step 2**

☐ Testing

Label

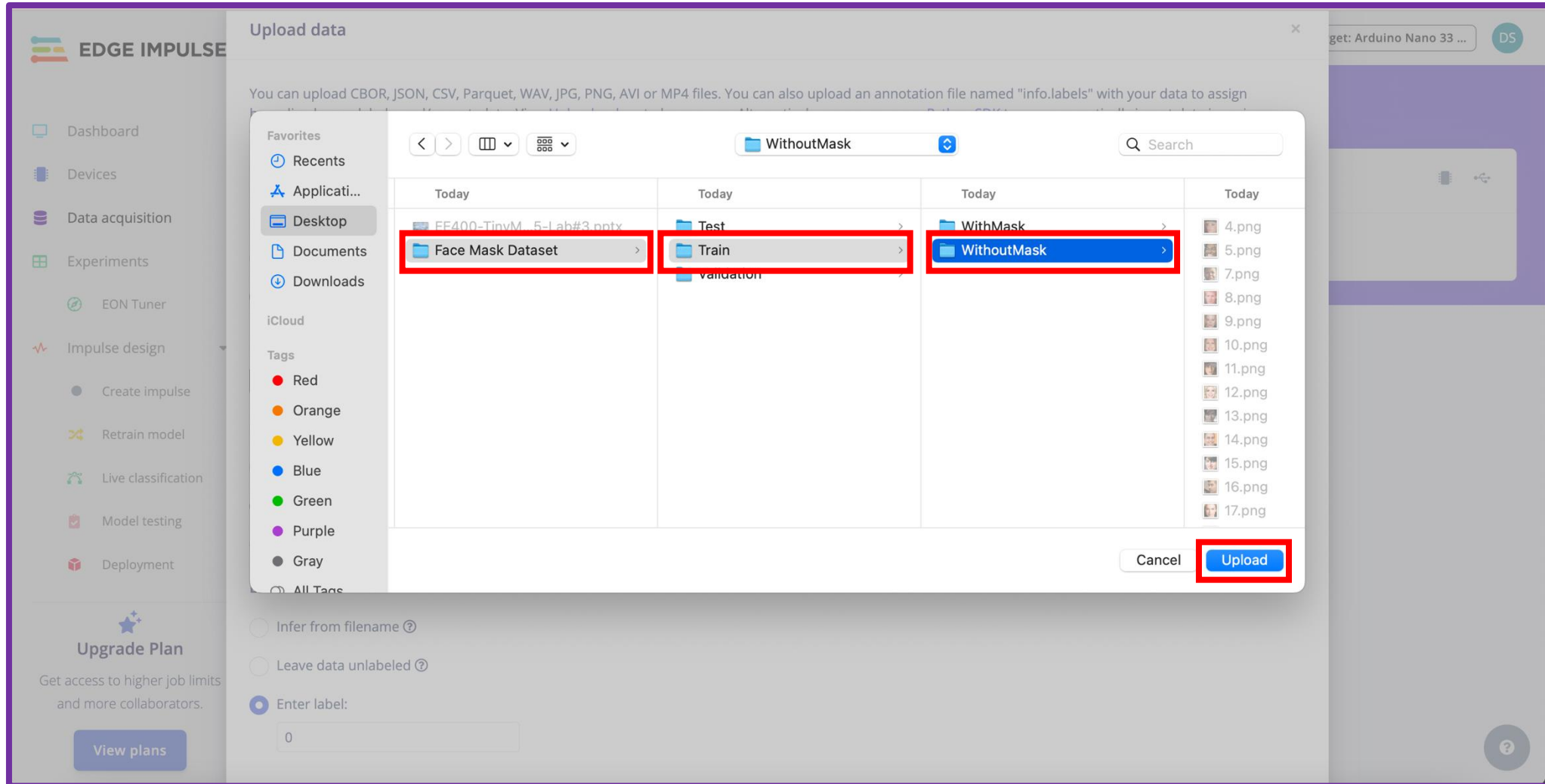
☐ Infer from filename ?

☐ Leave data unlabeled ?

☒ Enter label: **Step 3**

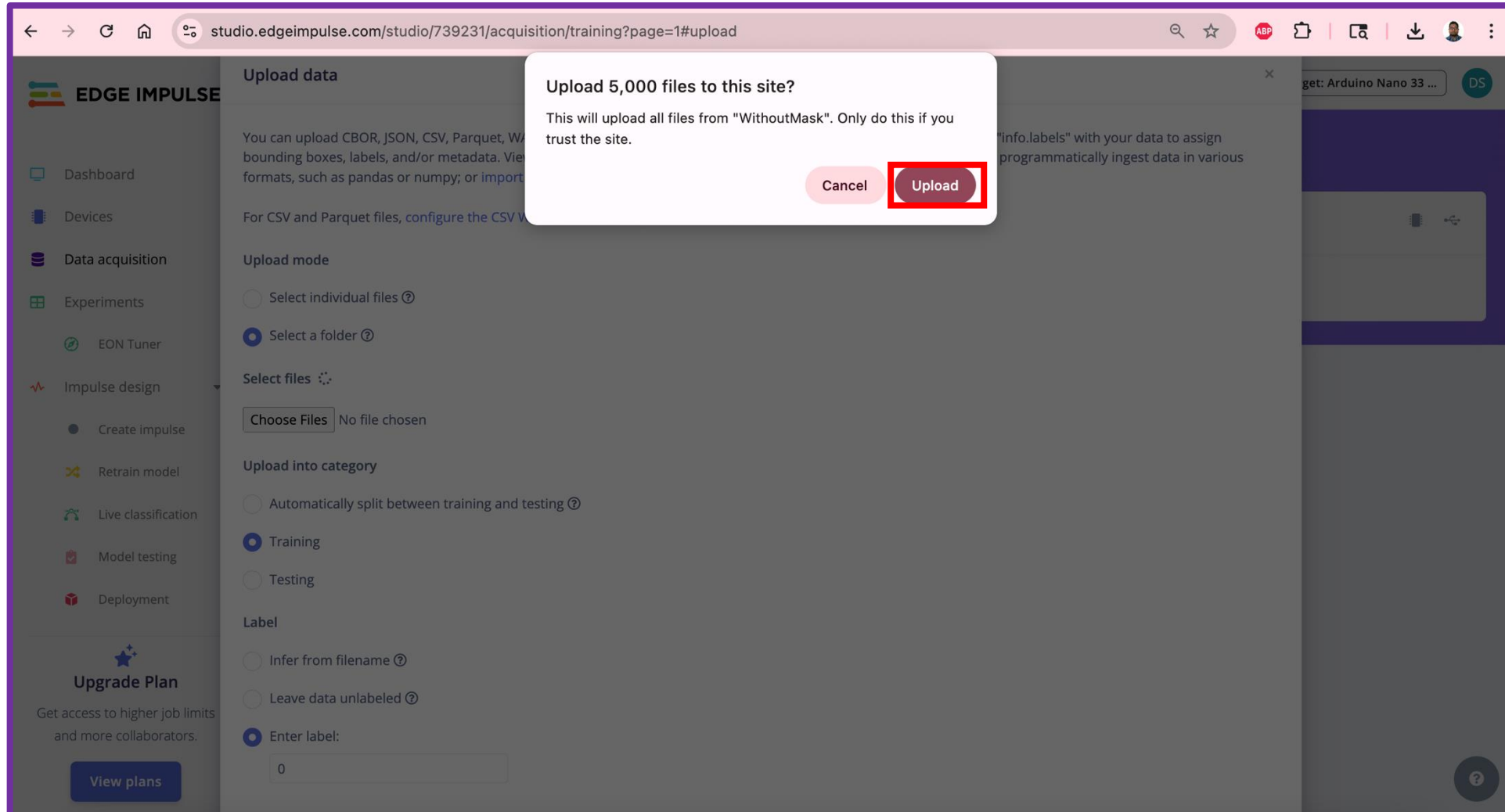
Lab 4 [Software] – Upload Training Data Without Mask

8. Upload the “Training Data Without Mask” into your Edge Impulse project



Lab 4 [Software] – Upload Training Data Without Mask

8. Upload the “Training Data Without Mask” into your Edge Impulse project



Lab 4 [Software] – Upload Training Data Without Mask

8. Upload the “Training Data Without Mask” into your Edge Impulse project

The screenshot displays the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Create impulse, Retrain model, Live classification, Model testing, and Deployment. Below these is an 'Upgrade Plan' section. The main content area shows the 'Upload data' workflow. It includes instructions on file formats and a link to the CSV Wizard. Under 'Upload mode', 'Select a folder' is selected. The 'Select files' section has a 'Choose Files' button and a file count of 5000. Under 'Upload into category', 'Training' is selected. The 'Label' section has 'Enter label:' selected, with a text input field containing the value '0'. At the bottom right, the 'Upload data' button is highlighted with a red rectangle. A 'Back' link is at the bottom left of the main area.

formats, such as pandas or numpy; or [import data directly from another project](#).

For CSV and Parquet files, [configure the CSV Wizard](#) to define how your files should be processed before uploading files.

Upload mode

☐ Select individual files ?

☒ Select a folder ?

Select files

[Choose Files](#) 5000 files

Upload into category

☐ Automatically split between training and testing ?

☒ Training

☐ Testing

Label

☐ Infer from filename ?

☐ Leave data unlabeled ?

☒ Enter label:

0

[< Back](#)

[Upload data](#)

Lab 4 [Software] – Upload Training Data Without Mask

8. Upload the “Training Data Without Mask” into your Edge Impulse project

EDGE IMPULSE

formats, such as pandas or numpy; or [import data directly from another project](#).

For CSV and Parquet files, [configure the CSV Wizard](#) to define how your files should be processed before uploading files.

Upload mode

- ☐ Select individual files ?
- ☒ Select a folder ?

Select files

[Choose Files](#) 5000 files

Upload into category

- ☐ Automatically split between training and testing
- ☒ Training
- ☐ Testing

Label

- ☐ Infer from filename ?
- ☐ Leave data unlabeled ?
- ☒ Enter label:

[Back](#)

Upload output

```
[ 578/5000] Uploading 4177.png OK
[ 579/5000] Uploading 5269.png OK
[ 580/5000] Uploading 3618.png OK
[ 581/5000] Uploading 4163.png OK
[ 582/5000] Uploading 2506.png OK
```

**We are doing
“Image Classification”
NOT Object Detection**

IMAGE DATA DETECTED!

Are you building an object
detection project?

You can change this choice under "Dashboard > Labeling method".

[Yes](#) [No](#)

[Upload data](#)

Lab 4 [Software] – Upload Training Data With Mask

9. Upload the “Training Data With Mask” into your Edge Impulse project Enter Label: “1”

The screenshot displays the Edge Impulse web interface for uploading training data. The left sidebar shows navigation options: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Create impulse, Retrain model, Live classification, Model testing, and Deployment. The main content area shows the upload process with the following steps highlighted:

- Step 1:** Select a folder (radio button selected).
- Step 2:** Choose Files (button highlighted).
- Step 3:** Training (radio button selected). A callout box indicates: “Face Mask Dataset” → “Train” → “WithMask”.
- Step 4:** Enter label: 1 (text input field highlighted).
- Step 5:** Upload data (button highlighted).

The right side of the interface shows the upload progress, listing files being uploaded (e.g., 586.png, 1774.png, etc.) and their status (OK). A message at the bottom states: “Done. Files uploaded successful: 5000. Files that failed to upload: 0. Job completed”.

Lab 4 [Software] – Upload Test Data Without Mask

10. Upload the “**Test Data Without Mask**” into your Edge Impulse project
Enter Label: “0”

The screenshot shows the Edge Impulse web interface with a sidebar on the left containing navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Create impulse, Retrain model, Live classification, Model testing, and Deployment. The main content area is titled "For CSV and Parquet files, configure the CSV Wizard to define how your files should be processed before uploading files." and contains the following sections:

- Upload mode:** Two radio buttons are present. "Select a folder" is selected and highlighted with a red box and labeled "Step 1".
- Select files:** A "Choose Files" button is highlighted with a red box and labeled "Step 2". It indicates "509 files" are selected.
- Upload into category:** Three radio buttons are present. "Testing" is selected and highlighted with a red box and labeled "Step 3". A purple callout box points to this section with the text: "Face Mask Dataset" → "Test" → "WithoutMask".
- Label:** The "Enter label:" option is selected and highlighted with a red box and labeled "Step 4". The input field contains the value "0".
- Upload data:** A blue button at the bottom right is highlighted with a red box and labeled "Step 5".

On the right side of the interface, there is a log titled "Upload output" showing a list of files being uploaded, each followed by "OK". At the bottom of the log, it states: "Done. Files uploaded successful: 4869. Files that failed to upload: 131."

Lab 4 [Software] – Upload Test Data With Mask

11. Upload the “Test Data With Mask” into your Edge Impulse project

Enter Label: “1”

The screenshot shows the Edge Impulse web interface with a sidebar on the left containing navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Upgrade Plan, and View plans. The main content area displays the 'Upload data' workflow. The steps are as follows:

- Step 1:** Under 'Upload mode', the 'Select a folder' radio button is selected and highlighted with a red box.
- Step 2:** Under 'Select files', the 'Choose Files' button is highlighted with a red box, and it shows '483 files'.
- Step 3:** Under 'Upload into category', the 'Testing' radio button is selected and highlighted with a red box. A purple callout box contains the text:
“Face Mask Dataset”
→ “Test”
→ “WithMask”
- Step 4:** Under 'Label', the 'Enter label:' option is selected and highlighted with a red box. The input field below it contains the number '1'.
- Step 5:** At the bottom right, the 'Upload data' button is highlighted with a red box.

The 'Upload output' section on the right shows a list of files being uploaded (e.g., 4795.png, 2482.png, etc.) with status 'OK'. At the bottom, it states: 'Done. Files uploaded successfully: 509. Files that failed to upload: 0. Job completed'.

Lab 4 [Software] – Observe Data

12. Observe uploaded data samples via “Data acquisition” tab in your Edge Impulse project

The screenshot shows the Edge Impulse project interface for a project named 'Mask Detection' by Dinuka Sahabandu. The 'Data acquisition' tab is selected in the left sidebar. The main area displays the 'Dataset' tab with a 'DATA COLLECTED' summary of 10,839 items and a 'TRAIN / TEST SPLIT' of 91% / 9%. A red box highlights the 'TRAIN / TEST SPLIT' section, and a blue callout box contains the text: 'Current Train/Test split is 91% / 9%. Next, we are going to change this to 80% / 20% by moving some samples from Training set to Test set'. Below this, a table lists dataset samples with columns for 'SAMPLE NAME', 'LABEL', and 'ADDED'. The sample 'Augmented_678_6388399' is highlighted in blue. To the right, a preview of the selected sample shows a person wearing a red face mask. The 'Metadata' section at the bottom indicates 'No metadata'.

EDGE IMPULSE

Dinuka Sahabandu / Mask Detection PERSONAL

Target: Arduino Nano 33 ... DS

Dataset Data explorer Data sources Synthetic data AI labeling CS Wizard

DATA COLLECTED 10,839 items

TRAIN / TEST SPLIT 91% / 9%

Dataset

Training (9,869) Test (970)

SAMPLE NAME	LABEL	ADDED
Augmented_383_82335	1	Today, 01:46:31
Augmented_650_1052629	1	Today, 01:46:31
Augmented_693_6192334	1	Today, 01:46:31
Augmented_363_2038261	1	Today, 01:46:31
Augmented_811_5146977	1	Today, 01:46:31
Augmented_491_5916085	1	Today, 01:46:31
Augmented_678_6388399	1	Today, 01:46:31
Augmented_50_771835	1	Today, 01:46:31

Augmented_678_6388399

Metadata ? +

No metadata.

Dashboard

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Impulse design

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Retrain model

Live classification

Model testing

Deployment

Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

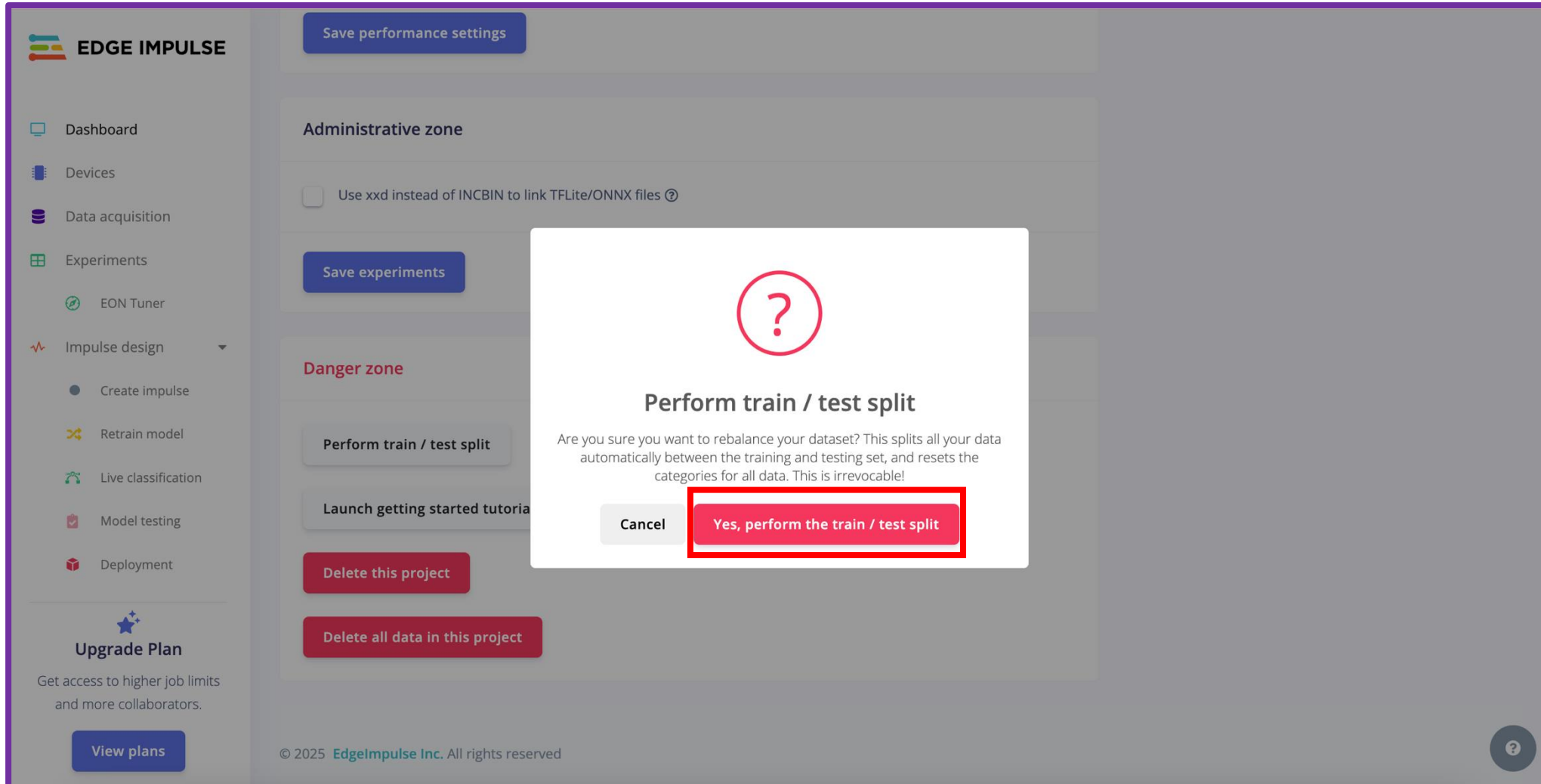
Lab 4 [Software] – Train/Test Split

13. Set Train/Test split via “Dashboard” tab in your Edge Impulse project

The screenshot shows the Edge Impulse web interface. On the left sidebar, the 'Dashboard' tab is highlighted with a red rectangle. The main content area is divided into sections. The 'Administrative zone' at the top contains a checkbox for 'Use xxd instead of INCBIN to link TFLite/ONNX files' and a 'Save performance settings' button. Below this is the 'Danger zone' section, which contains a button labeled 'Perform train / test split' (highlighted with a red rectangle), a 'Launch getting started tutorial' button, and two red buttons: 'Delete this project' and 'Delete all data in this project'. A blue callout box with a wavy border points to the 'Perform train / test split' button and contains the text 'Scroll down to “Danger zone” section'. At the bottom of the sidebar, there is an 'Upgrade Plan' section with a 'View plans' button. The footer of the interface shows the copyright notice '© 2025 EdgeImpulse Inc. All rights reserved' and a help icon.

Lab 4 [Software] – Train/Test Split

13. Set Train/Test split via “Dashboard” tab in your Edge Impulse project



Lab 4 [Software] – Train/Test Split

14. Observe updated Train/Test split via “Data acquisition” tab in your Edge Impulse project

The screenshot shows the Edge Impulse web interface for a project named 'Mask Detection' by 'Dinuka Sahabandu'. The 'Data acquisition' tab is selected in the left sidebar. The main content area shows a dataset with 10,839 items and a train/test split of 80% / 20%. A table lists training samples with their names, labels, and addition times.

SAMPLE NAME	LABEL	ADDED
Augmented_797_8118850	1	Today, 01:50:09
Augmented_673_1693056	1	Today, 01:50:09
Augmented_677_9933265	1	Today, 01:50:09
Augmented_744_2437969	1	Today, 01:50:09
351	1	Today, 01:50:09
423	1	Today, 01:50:08
Augmented_569_2915532	1	Today, 01:50:08
Augmented_164_2235307	1	Today, 01:50:08

Lab 4 [Software] – Create Impulse

15. Setup the ML Pipeline via “Create Impulse” tab in your Edge Impulse project

The screenshot displays the Edge Impulse web interface for a project named 'Mask Detection' by 'Dinuka Sahabandu'. The interface is divided into a left sidebar and a main workspace. The sidebar contains navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Retrain model, Live classification, Model testing, and Deployment. The 'Impulse design' section is expanded, showing a 'Create impulse' button highlighted with a red box and labeled 'Step 1'. The main workspace shows the 'Impulse #1' configuration. It includes a description: 'An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.' Below this, there are three main blocks: 'Image data' (red), 'Add a processing block' (light blue, highlighted with a red box and labeled 'Step 3'), and 'Add a learning block' (light blue). The 'Image data' block is further detailed with 'Input axes' set to 'image', 'Image width' and 'Image height' both set to '48' (highlighted with a blue box and labeled 'Step 2'), and 'Resize mode' set to 'Fit shortest axis'. A red arrow points from the 'Image width' and 'Image height' fields to a callout box that says 'Set Image width and height to 48 x 48'. To the right of the 'Add a learning block' block is an 'Output features' block (green) with a checkmark icon. A 'Save Impulse' button is located at the bottom right of the workspace. The footer of the interface shows '© 2025 EdgeImpulse Inc. All rights reserved' and a 'View plans' button.

Step 1

Step 2

Step 3

Set Image width and height to 48 × 48

Lab 4 [Software] – Create Impulse

16. Add “Image” data processing block

The screenshot shows the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Retrain model, Live classification, Model testing, and Deployment. The main area displays 'Impulse #1' with a red 'Image data' block. A modal dialog titled 'Add a processing block' is open in the center. It contains a table with two blocks: 'Image' and 'Raw Data'. The 'Image' block is highlighted with a red rectangle. The 'Image' block has the description 'Preprocess and normalize image data, and optionally reduce the color depth.' and is marked as 'OFFICIALLY SUPPORTED'. The 'Raw Data' block has the description 'Use data without pre-processing. Useful if you want to use deep learning to learn features.' and is also marked as 'OFFICIALLY SUPPORTED'. Both blocks are by 'Edge Impulse' and have a star icon. There are 'Add' buttons next to each block. At the bottom of the dialog, there is a link 'Add custom block' and a 'Cancel' button. The background interface shows an 'Input axes' section with 'Image width' set to 48 and a 'Resize mode' dropdown set to 'Fit shortest'. There is also an 'Output features' section with a checkmark and a 'Save Impulse' button.

DESCRIPTION	AUTHOR	RECOMMENDED
Image OFFICIALLY SUPPORTED Preprocess and normalize image data, and optionally reduce the color depth.	Edge Impulse	★ <button>Add</button>
Raw Data OFFICIALLY SUPPORTED Use data without pre-processing. Useful if you want to use deep learning to learn features.	Edge Impulse	<button>Add</button>

Some processing blocks have been hidden based on the data in your project. [Show all blocks anyway](#)

[Add custom block](#) Cancel

Lab 4 [Software] – Create Impulse

17. Click on “Add a learning block”

The screenshot displays the Edge Impulse web interface for configuring 'Impulse #1'. The interface is divided into several sections:

- Left Sidebar:** Contains navigation links for Dashboard, Devices, Data acquisition, Experiments, EON Tuner, and Impulse design. Under 'Impulse design', there are options for 'Create impulse', 'Retrain model', 'Live classification', 'Model testing', and 'Deployment'. At the bottom, there is an 'Upgrade Plan' section with a 'View plans' button.
- Top Header:** Labeled 'Impulse #1', it includes a description: 'An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.'
- Main Configuration Area:**
 - Image data (Red box):** Includes 'Input axes' set to 'image', 'Image width' and 'Image height' both set to 48, and 'Resize mode' set to 'Fit shortest axis'.
 - Image (White box):** Includes a 'Name' field set to 'Image', 'Input axes (1)' set to 'Image', and a dropdown menu set to 'image'.
 - Add a learning block (Red box):** A button with a flask icon and the text 'Add a learning block', which is highlighted with a red border.
 - Output features (Teal box):** Includes a checkmark icon and a 'Save Impulse' button.
 - Add a processing block (White box):** A button with a lightning bolt icon and the text 'Add a processing block'.

Lab 4 [Software] – Create Impulse

18. Add “Transfer Learning (Images)” as your learning block

The screenshot shows the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Retrain model, Live classification, Model testing, and Deployment. The main area displays 'Impulse #1' with a red sidebar for 'Image data' containing 'Input axes' (image), 'Image width' (48), and 'Resize mode' (Fit shortest). A modal window titled 'Add a learning block' is open in the center. It contains a table of learning blocks:

DESCRIPTION	AUTHOR	RECOMMENDED
Classification OFFICIALLY SUPPORTED Learns patterns from data, and can apply these to new data. Great for categorizing movement or recognizing audio.	Edge Impulse	<button>Add</button>
Transfer Learning (Images) OFFICIALLY SUPPORTED Fine tune a pre-trained image classification model on your data. Good performance even with relatively small image datasets.	Edge Impulse	<button>Add</button>
Regression OFFICIALLY SUPPORTED Learns patterns from data, and can apply these to new data. Great for predicting numeric continuous values.	Edge Impulse	<button>Add</button>
Visual Anomaly Detection - FOMO-AD OFFICIALLY SUPPORTED Detect visual anomalies. Extracts visual features using a pre-trained backbone, and applies a scoring function to evaluate how anomalous a sample is by comparing the extracted features to the learned model. Does not require anomalous data.	Edge Impulse	<button>Add</button>

Below the table is a lightning bolt icon and the text: 'Want access to all learning blocks? Upgrade now.' At the bottom of the modal, it says: 'Some learning blocks have been hidden based on the data in your project. Show all blocks anyway'. In the background, the 'Output features' section is visible with a green box and a checkmark, and a 'Save Impulse' button.

Lab 4 [Software] – Save Impulse

19. Click on “Save Impulse” to save your ML Pipeline

The screenshot displays the Edge Impulse web interface for configuring an ML pipeline. The interface is divided into a left sidebar, a top header, and a main workspace.

Left Sidebar: Contains navigation links for Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Retrain model, Live classification, Model testing, and Deployment. At the bottom, there is an "Upgrade Plan" section with a "View plans" button.

Top Header: Labeled "Impulse #1", it includes a description: "An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data."

Main Workspace: The workspace contains four colored blocks representing different stages of the pipeline:

- Image data (Red block):** Configured with "Input axes" set to "image", "Image width" and "Image height" both set to 48, and "Resize mode" set to "Fit shortest axis".
- Image (White block):** Configured with "Name" set to "Image" and "Input axes (1)" set to "image".
- Transfer Learning (Images) (Blue block):** Configured with "Name" set to "Transfer learning", "Input features" checked for "Image", and "Output features" set to "2 (0, 1)".
- Output features (Green block):** Configured with "Output features" set to "2 (0, 1)".

A red rectangular box highlights the "Save Impulse" button located in the bottom right corner of the workspace. Below the main blocks, there are two dashed boxes labeled "Add a processing block" and "Add a learning block". A help icon (?) is visible in the bottom right corner of the workspace.

Lab 4 [Software] – Setup Image Data

20. Setup the image data via “Image” tab in your Edge Impulse project

The screenshot displays the Edge Impulse web interface. On the left sidebar, the 'Image' option is selected and highlighted with a red box. The top navigation bar shows the 'Generate features' button highlighted with a red box. The main content area is divided into several sections:

- Raw data:** Displays a sample image of a person wearing a green face mask. It includes filters for 'Show: All labels' and 'Augmented_797_811'.
- Raw features:** Shows a list of hexadecimal values representing raw image data: 0x5a5d52, 0x5a5d52, 0x585b51, 0x575a50, 0x56594f, 0x55584f, 0x55584f, 0x55584f...
- Parameters:** Includes a section for 'Image' with a 'Color depth' dropdown menu set to 'RGB'. A 'Save parameters' button is located at the bottom of this section.
- DSP result:** Displays the processed image of the person wearing the green face mask.
- Processed features:** Shows a list of numerical values representing processed image data: 0.3529, 0.3647, 0.3216, 0.3529, 0.3647, 0.3216, 0.3451, 0.3569, 0.3176, 0.3412...
- On-device performance:** A section with a question mark icon, indicating performance metrics.

Lab 4 [Software] – Setup Image Data

20. Setup the image data via “Image” tab in your Edge Impulse project

The screenshot displays the Edge Impulse web interface for a project named "Mask Detection" by Dinuka Sahabandu. The interface is divided into a left sidebar, a main content area, and a top header. The sidebar contains navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Image, Transfer learning, Retrain model, and Live classification. The main content area has two tabs: "Parameters" and "Generate features" (active). Under "Parameters", there is a "Training set" section with a table showing "Data in training set" as 8,622 items and "Classes" as 2 (0, 1). Below this table is a blue "Generate features" button, which is highlighted with a red rectangle. To the right of the "Training set" section is a "Feature explorer" section with the text "No features generated yet." Below the "Training set" section is a "Feature generation output" section showing 0 items. The top header includes the user name "Dinuka Sahabandu", the project name "Mask Detection", a "PERSONAL" label, and a "Target: Arduino Nano 33 ..." dropdown. A "DS" badge is also visible in the top right corner. The footer of the interface shows the copyright "© 2025 EdgeImpulse Inc. All rights reserved" and a "View plans" button.

EDGE IMPULSE

Dinuka Sahabandu / Mask Detection PERSONAL Target: Arduino Nano 33 ... DS

Parameters **Generate features**

Training set

Data in training set	8,622 items
Classes	2 (0, 1)

Generate features

Feature explorer

No features generated yet.

Feature generation output (0)

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Upgrade Plan
Get access to higher job limits and more collaborators.
[View plans](#)

Lab 4 [Software] – Transfer Learning

21. Setup ML model via “Transfer Learning” tab in your Edge Impulse project

The screenshot displays the Edge Impulse web interface. On the left is a sidebar with navigation options: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Image, **Transfer learning** (highlighted with a red box), Retrain model, and Live classification. Below the sidebar is an 'Upgrade Plan' section. The main content area is titled 'Neural Network settings' and includes sections for 'Training settings' (with fields for Number of training cycles: 20, Use learned optimizer: unchecked, Learning rate: 0.0005, Training processor: CPU, and Data augmentation: unchecked) and 'Advanced training settings'. The 'Neural network architecture' section shows an 'Input layer (27,648 features)' and a 'MobileNetV2 96x96 0.35 (final layer: 16 neurons, 0.1 dropou)' model. Below the model is a red box containing the text 'Choose a different model'. At the bottom is an 'Output layer (2 classes)'. On the right side of the interface is a 'Training output' section with a filter icon and '(0)'.

Lab 4 [Software] – Transfer Learning

21. Setup ML model via “Transfer Learning” tab in your Edge Impulse project

MobileNetV1 96x96 0.2 OFFICIALLY SUPPORTED Uses around 83.1K RAM and 218.3K ROM with default settings and optimizations. Works best with 96x96 input size. Supports both RGB and grayscale.	Edge Impulse	Add
MobileNetV1 96x96 0.1 OFFICIALLY SUPPORTED Uses around 53.2K RAM and 101K ROM with default settings and optimizations. Works best with 96x96 input size. Supports both RGB and grayscale.	Edge Impulse	Add
MobileNetV2 96x96 0.35 OFFICIALLY SUPPORTED Uses around 296.8K RAM and 575.2K ROM with default settings and optimizations. Works best with 96x96 input size. Supports both RGB and grayscale.	Edge Impulse	Add
MobileNetV2 96x96 0.1 OFFICIALLY SUPPORTED Uses around 270.2K RAM and 212.3K ROM with default settings and optimizations. Works best with 96x96 input size. Supports both RGB and grayscale.	Edge Impulse	Add
MobileNetV2 96x96 0.05 OFFICIALLY SUPPORTED Uses around 265.3K RAM and 162.4K ROM with default settings and optimizations. Works best with 96x96 input size. Supports both RGB and grayscale.	Edge Impulse	Add
MobileNetV2 160x160 1.0 OFFICIALLY SUPPORTED Uses around 1.3M RAM and 2.6M ROM with default settings and optimizations. Works best with 160x160 input size. Supports RGB only.	Edge Impulse	Add
MobileNetV2 160x160 0.75 OFFICIALLY SUPPORTED Uses around 1.3M RAM and 1.7M ROM with default settings and optimizations. Works best with 160x160 input size. Supports RGB only.	Edge Impulse	Add
MobileNetV2 160x160 0.5 OFFICIALLY SUPPORTED Uses around 700.7K RAM and 982.4K ROM with default settings and optimizations. Works best with 160x160 input size. Supports RGB only.	Edge Impulse	Add

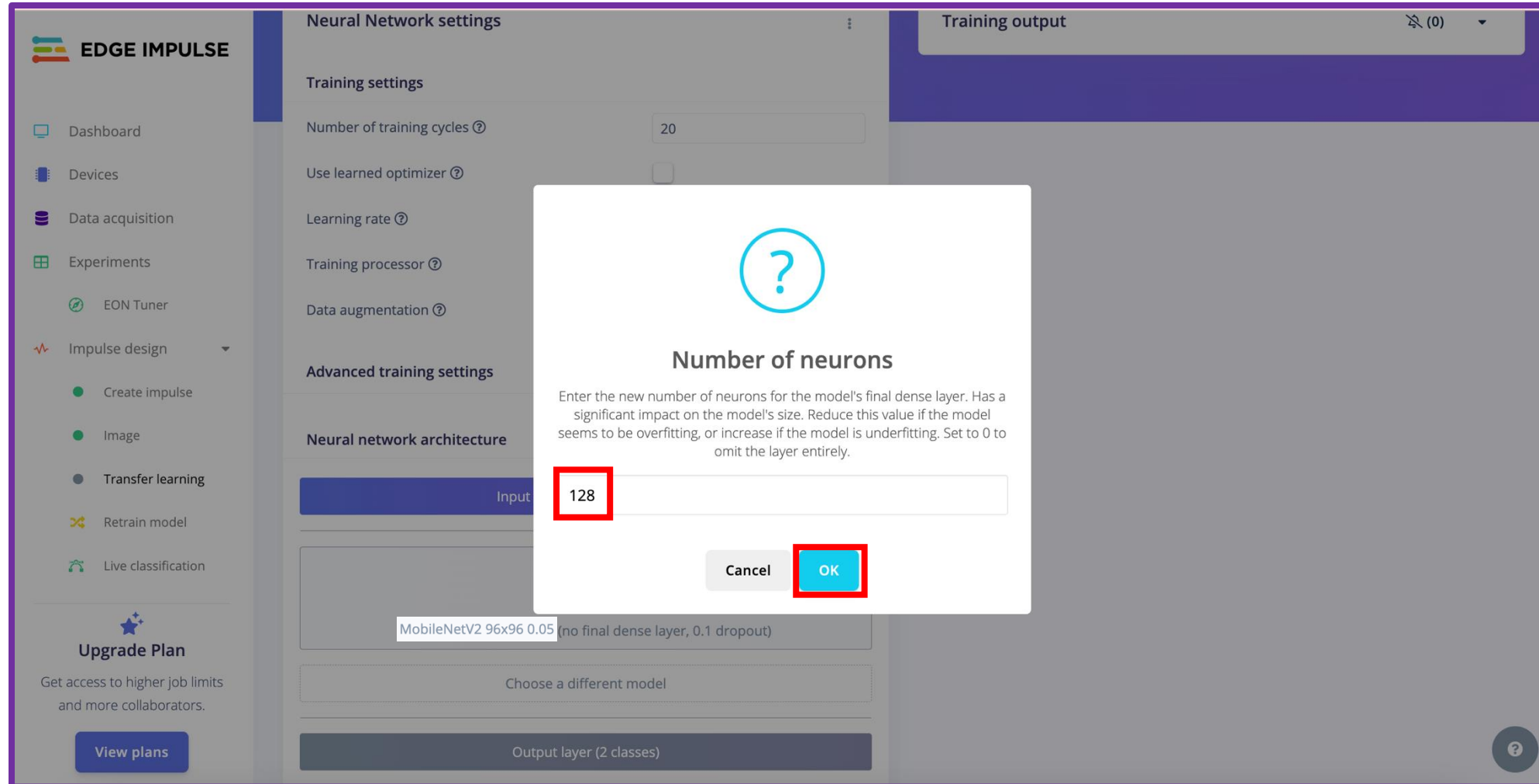
Lab 4 [Software] – Transfer Learning

21. Setup ML model via “Transfer Learning” tab in your Edge Impulse project

The screenshot displays the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Image, Transfer learning (highlighted), Retrain model, and Live classification. Below these is an 'Upgrade Plan' section. The main content area is titled 'Neural Network settings' and includes 'Training settings' (Number of training cycles: 20, Use learned optimizer: unchecked, Learning rate: 0.0005, Training processor: CPU, Data augmentation: unchecked) and 'Advanced training settings' (collapsed). Under 'Neural network architecture', there is an 'Input layer (27,648 features)' and a highlighted 'MobileNetV2 96x96 0.05 (no final dense layer, 0.1 dropout)' layer. A red box and a red hexagon highlight the MobileNetV2 layer. Below it is a 'Choose a different model' button, and at the bottom is an 'Output layer (2 classes)'. On the right, a 'Training output' section shows '(0)' results.

Lab 4 [Software] – Transfer Learning

21. Setup ML model via “Transfer Learning” tab in your Edge Impulse project



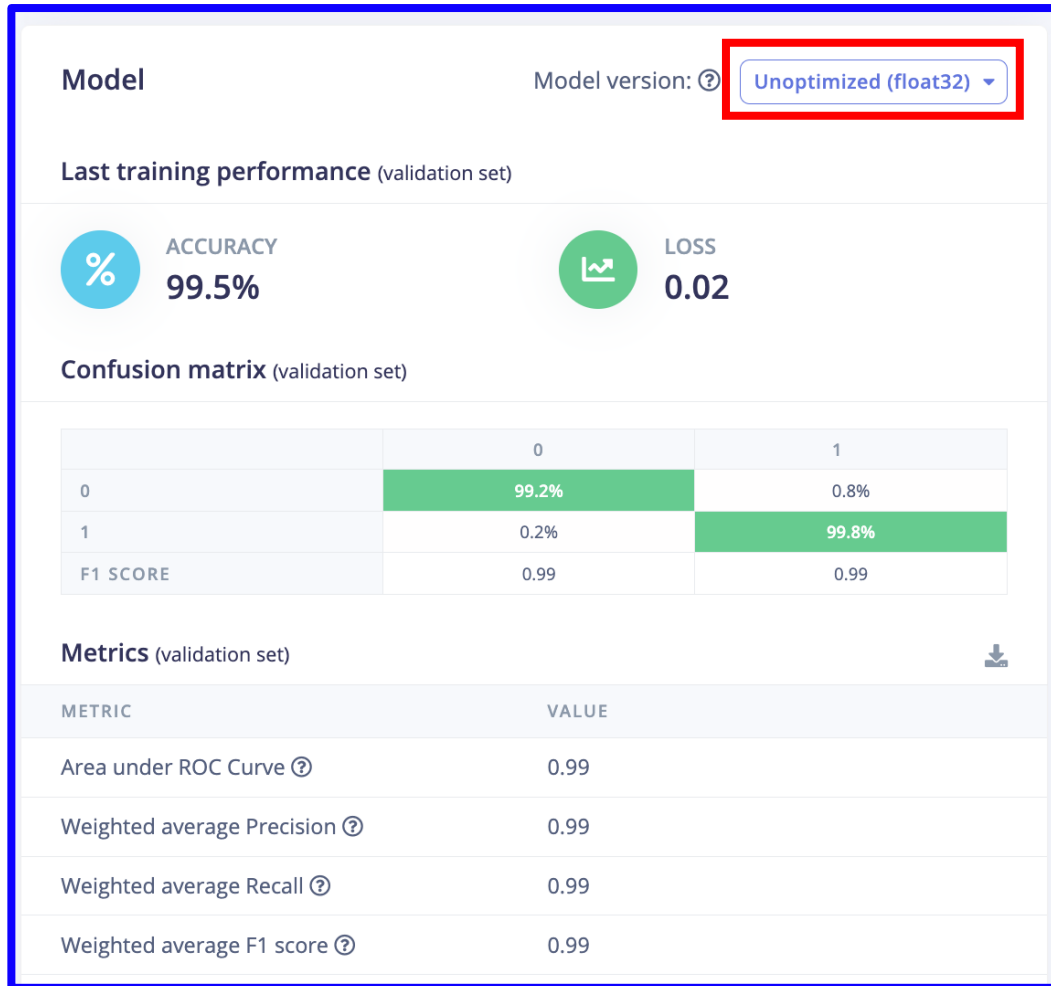
Lab 4 [Software] – Training via Transfer Learning

22. Select “Save & Train” to perform transfer learning on your ML model

The screenshot displays the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Image, Transfer learning, Retrain model, and Live classification. Below these is an 'Upgrade Plan' section. The main area shows training configuration: 'Number of training cycles' is 20, 'Use learned optimizer' is unchecked, 'Learning rate' is 0.0005, 'Training processor' is CPU, and 'Data augmentation' is unchecked. An 'Advanced training settings' dropdown is visible. Below this, the 'Neural network architecture' section shows an 'Input layer (27,648 features)', a central box for 'MobileNetV2 96x96 0.05 (final layer: 128 neurons, 0.1 dropout)' with a 'Choose a different model' link, and an 'Output layer (2 classes)'. At the bottom right, the 'Save & train' button is highlighted with a red rectangle. A help icon (?) is in the bottom right corner.

Lab 4 [Software] – Training Performance on Base Model

23. Observe the training performance of your base (unoptimized) model



Lab 4 [Software] – Training Performance on Compressed Model

24. Observe the training performance of your compressed (int8 quantized/optimized) model

ModelModel version: ③ Quantized (int8) ▾

Last training performance (validation set)

 **ACCURACY**
99.2%

 **LOSS**
0.04

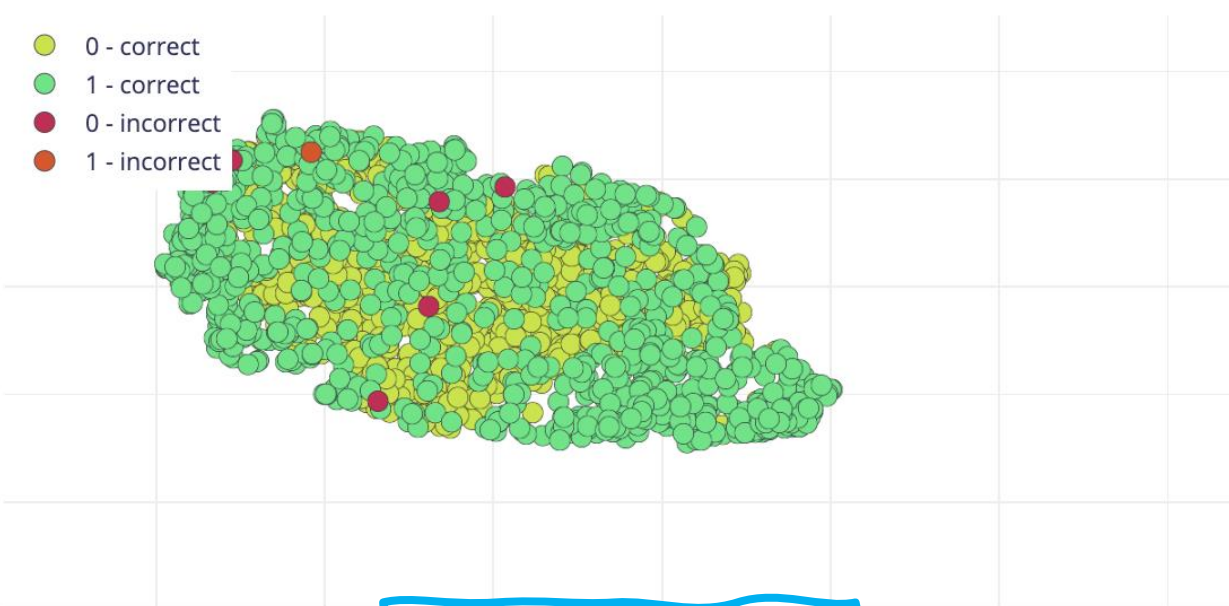
Confusion matrix (validation set)

	0	1
0	98.9%	1.1%
1	0.4%	99.6%
F1 SCORE	0.99	0.99

Metrics (validation set) 

METRIC	VALUE
Area under ROC Curve ③	0.99
Weighted average Precision ③	0.99
Weighted average Recall ③	0.99
Weighted average F1 score ③	0.99

Data explorer (full training set) ③



On-device performance ③ TensorFlow Lite™ Compiler (RAM optimized) ▾

 **INFERRNCING TI...**
829 ms.

 **PEAK RAM USAGE**
134.3K

 **FLASH USAGE**
328.8K

This is much better!
(Less than 250K)

Lab 4 [Software] – Test Performance

25. Evaluate your unoptimized and optimized models on test data (data not used during the training – unseen data) via “**Model Testing**” tab in your Edge Impulse project

The screenshot displays the Edge Impulse web interface for a project named "Mask Detection" by Dinuka Sahabandu. The left sidebar contains navigation options: Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Create impulse, Image, Transfer learning, Retrain model, Live classification, **Model testing** (highlighted with a red box), Deployment, and Versioning. The main content area is divided into two panels. The left panel, titled "Test data", contains a "Classify all" button (highlighted with a red box) and a table of test data. The right panel, titled "Model testing output", shows a filter icon and "(0)".

Target: Arduino Nano 33 ... DS

This lists all test data. You can manage this data through [Data acquisition](#).

Test data Classify all

Set the 'expected outcome' for each sample to the desired outcome to automatically score the impulse.

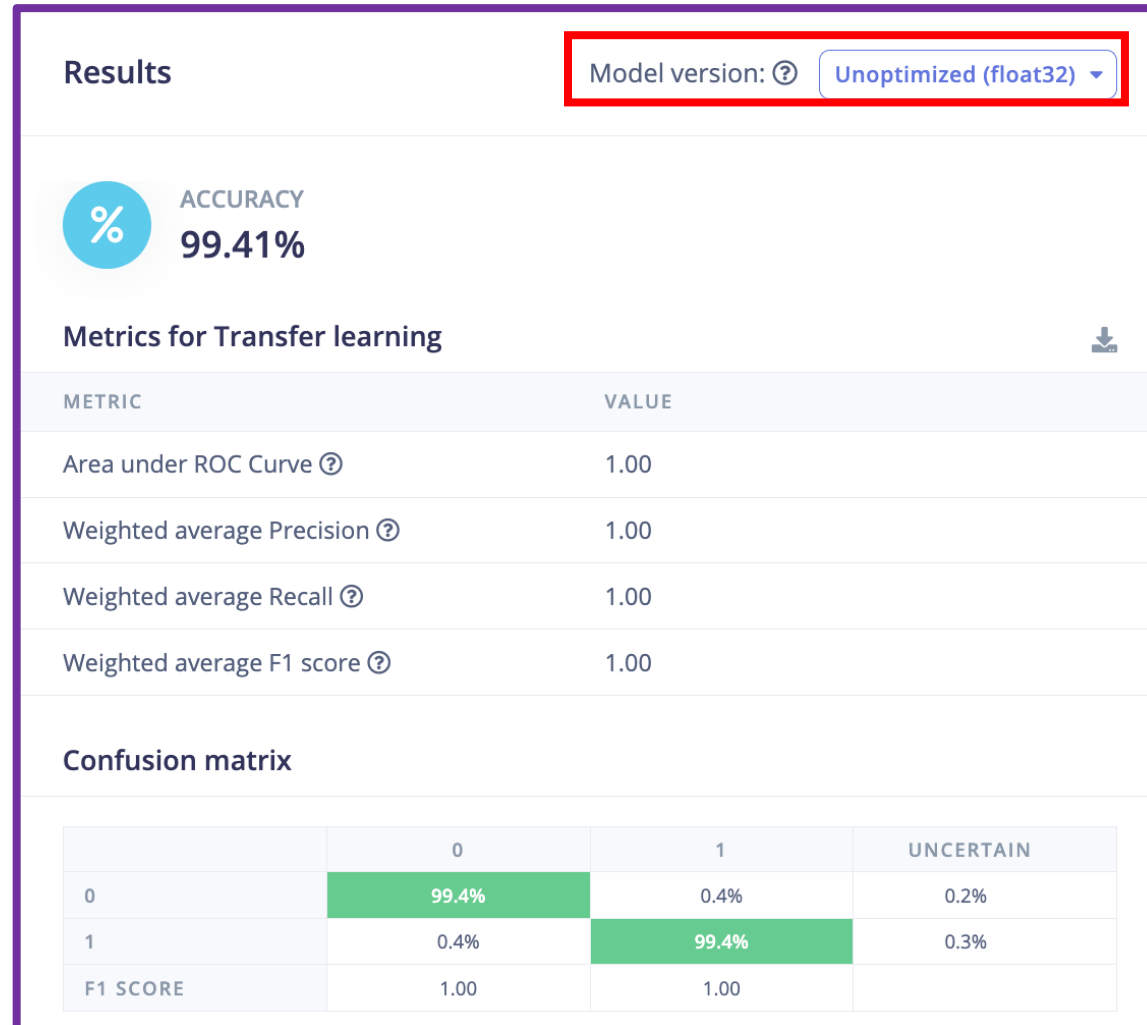
SAMPLE NAME	EXPECTED OUTCOME	ACCURACY	RESULT
Augmented_7...	1		...
Augmented_1...	1		...
Augmented_3...	1		...
Augmented_6...	1		...
Augmented_1...	1		...
Augmented_6...	1		...
Augmented_2...	1		...
Augmented_7...	1		...

Model testing output 🔍 (0)

Upgrade Plan
Get access to higher job limits and more collaborators.
[View plans](#)

Lab 4 [Software] – Test Performance on Base Model

23. Observe the test performance (on unseen data) of your base (unoptimized) model



Lab 4 [Software] – Training Performance on Compressed Model

24. Observe the test performance (on unseen data) of your compressed (int8 quantized/optimized) model

The screenshot displays the AWS SageMaker console interface. On the left is a navigation sidebar with options: Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Create impulse, Image, Transfer learning, Retrain model, Live classification, Model testing, Deployment, and Versioning. The main area is titled 'Test data' and contains a table with columns: SAMPLE NAME, EXPECTED OUTCOME, ACCURACY, and a dropdown menu. The table lists several 'Augmented' samples, all showing 100% accuracy. A red box highlights the 'Classify all' button and the gear icon. A blue box highlights the 'Test 'Quantized (int8)' model' option in the dropdown menu. A white box labeled 'Step 2' is positioned over the table. A blue box with text 'Click the gear icon and re-run "Classify all" to view results for your int8 mode' is also present. The right side of the console shows the 'Results' section for the 'Unoptimized (float32)' model, displaying an accuracy of 99.41% and a table of metrics for Transfer learning.

Test data

Step 3 **Classify all** ⚙️ Step 1 Testing output

Set thresholds

MODEL VARIANTS

Test 'Quantized (int8)' model

Step 2

Click the gear icon and re-run "Classify all" to view results for your int8 mode

Results Model version: Unoptimized (float32)

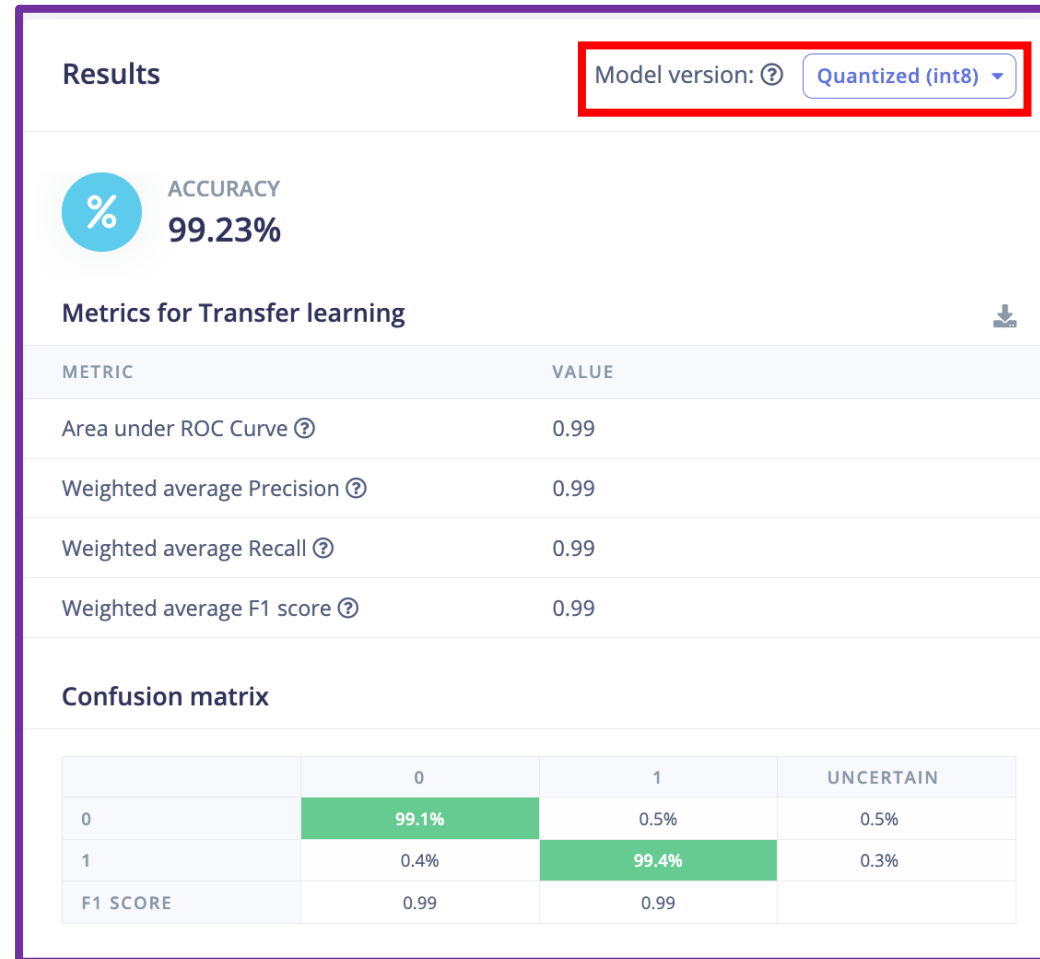
ACCURACY 99.41%

Metrics for Transfer learning

METRIC	VALUE
Area under ROC Curve	1.00
Weighted average Precision	1.00
Weighted average Recall	1.00
Weighted average F1 score	1.00

Lab 4 [Software] – Training Performance on Compressed Model

25. Observe the test performance (on unseen data) of your compressed (int8 quantized/optimized) model



Lab 4 [Software] – Model Deployment

26. Deploying your int8 model via “Deployment” tab in your Edge Impulse project

The screenshot shows the Edge Impulse web interface. On the left sidebar, the 'Deployment' tab is highlighted with a red box. The main content area is titled 'Configure your deployment' and includes a search bar where 'Arduino Nano 33 BLE Sense' is entered. Below the search bar, a red box highlights the search result for 'Arduino Nano 33 BLE Sense', which is described as 'A binary containing both the Edge Impulse data acquisition client and your full impulse.' Below this, there is a section for 'MODEL OPTIMIZATIONS' and a table showing performance metrics for the 'Quantized (int8)' model.

Configure your deployment

You can deploy your impulse to any device. This makes the model run without an internet connection, minimizes latency, and runs with minimal power consumption. [Read more.](#)

Search results:

- Arduino Nano 33 BLE Sense**
A binary containing both the Edge Impulse data acquisition client and your full impulse.
- EON™ Compiler**
A portable C++ library with no external dependencies, which can be compiled with any modern C++ compiler.

MODEL OPTIMIZATIONS

Model optimizations can increase on-device performance but may reduce accuracy.

Quantized (int8)

Selected ✓

	IMAGE	TRANSFER LEARN...	TOTAL
LATENCY	11 ms.	829 ms.	840 ms.
RAM	4.0K	125.0K	125.0K
FLASH	-	316.9K	-
ACCURACY			99.23%

Run this model

Scan QR code or launch in browser to test your prototype



[Launch in browser](#)

Lab 4 [Software] – Model Deployment

27. Select “Build” to generate binaries of the int8 model for deployment

The screenshot displays the EON Tuner interface. On the left is a sidebar with navigation options: EON Tuner, Impulse design, Create impulse, Image, Transfer learning, Retrain model, Live classification, Model testing, Deployment, Versioning, GETTING STARTED, Documentation, Forums, Upgrade Plan, and View plans. The main area is titled 'MODEL OPTIMIZATIONS' and includes a warning: 'Model optimizations can increase on-device performance but may reduce accuracy.' Below this, the 'EON™ Compiler' is selected, showing 'Same accuracy, 18% less RAM, 19% less ROM.' Two tables compare 'Quantized (int8)' and 'Unoptimized (float32)' models. The 'Quantized (int8)' model is selected. A large blue 'Build' button is at the bottom.

MODEL OPTIMIZATIONS
Model optimizations can increase on-device performance but may reduce accuracy.

EON™ Compiler
Same accuracy, 18% less RAM, 19% less ROM.

Quantized (int8)
Selected ✓

	IMAGE	TRANSFER LEARN...	TOTAL
LATENCY	11 ms.	829 ms.	840 ms.
RAM	4.0K	125.0K	125.0K
FLASH	-	316.9K	-
ACCURACY			99.23%

Unoptimized (float32)
Select

	IMAGE	TRANSFER LEARN...	TOTAL
LATENCY	11 ms.	8,208 ms.	8,219 ms.
RAM	4.0K	300.3K	300.3K
FLASH	-	985.7K	-
ACCURACY			99.41%

Estimate for Arduino Nano 33 BLE Sense (Cortex-M4F 64MHz).

Build

Lab 4 [Software] – Model Deployment

28. Upon successful model build you will see the following. You should be now able to see **“.zip”** file in your **“Downloads”** folder that contain **flash files** for model deployment. **Unzip this folder.**

Built Arduino Nano 33 BLE Sense firmware

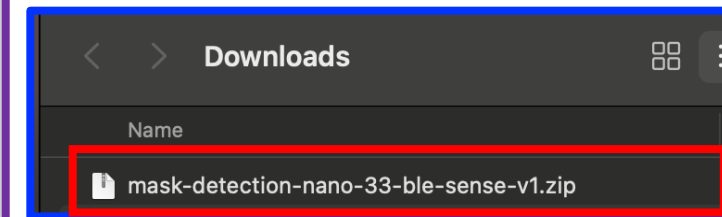
Click on the script for your operating system to flash the binary.

flash_mac.command

```
To update your account to use zsh, please run 'chsh -s /bin/zsh'.
For more details, please visit https://support.apple.com/kb/HT208050.
janjongboom@mac: ~ % /Users/janjongboom/Downloads/ei-continuous-gestures-deployment-a
rduno-nano-33-ble-sense-1590146532746/flash_mac.command ; exit;
Finding Arduino Mbed core...
Finding Arduino Nano 33 BLE...
Finding Arduino Nano 33 BLE OK
Flashing board...
Device      : nRF52840-QIAA
Version    : Arduino Bootloader (SAM-BA extended) 2.0 (Arduino:IKXYZ)
Address     : 0x0
Pages      : 256
Page Size  : 4096 bytes
Total Size : 1024KB
Planes     : 1
Lock Regions: 0
Locked     : none
Security    : false
Erase flash
Done in 0.000 seconds
Write 527696 bytes to flash (129 pages)
[=====] 23% (30/129 pages)
```

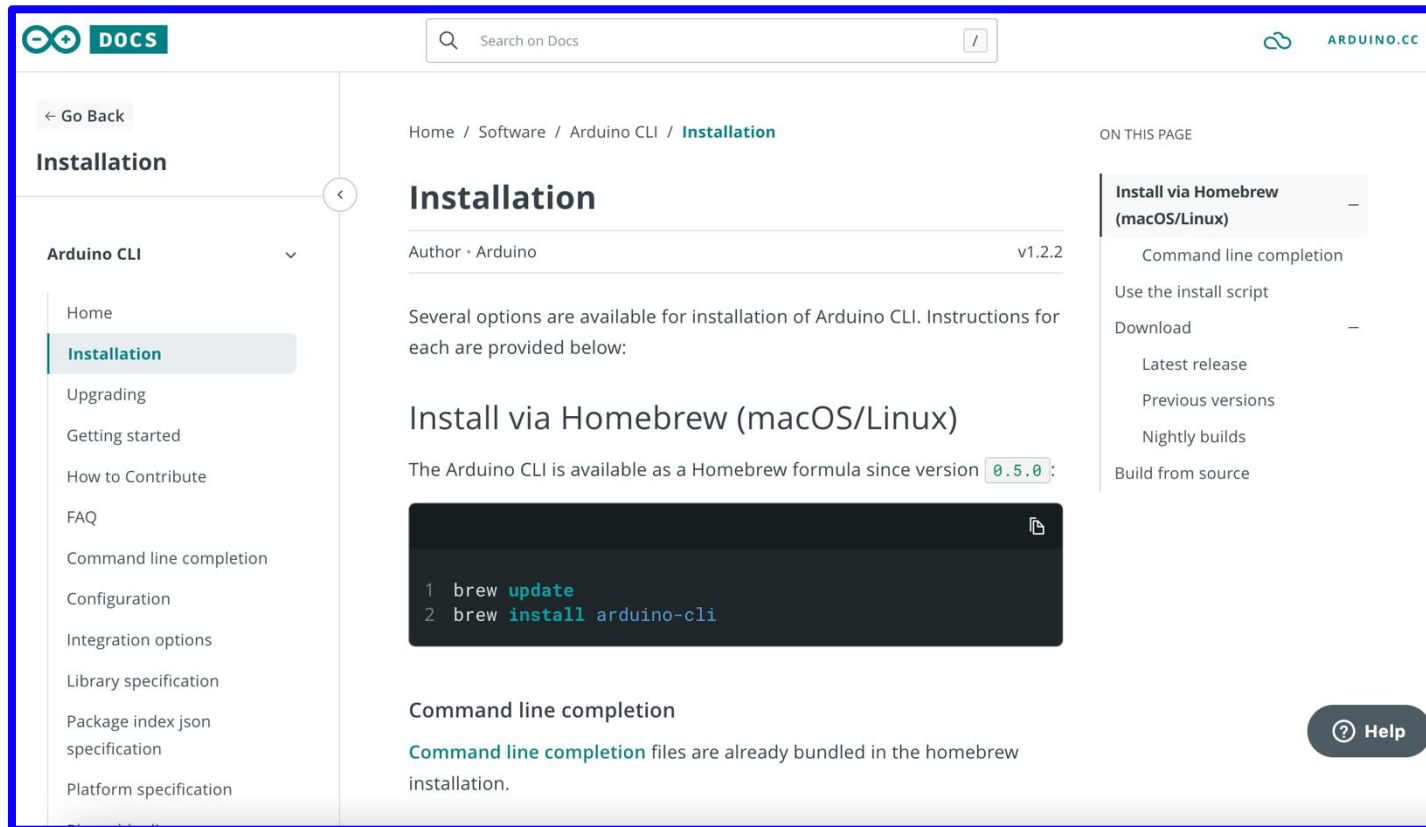
Try live inference preview on your device, from your terminal or command prompt run: `edge-impulse-run-impulse --debug`

Build



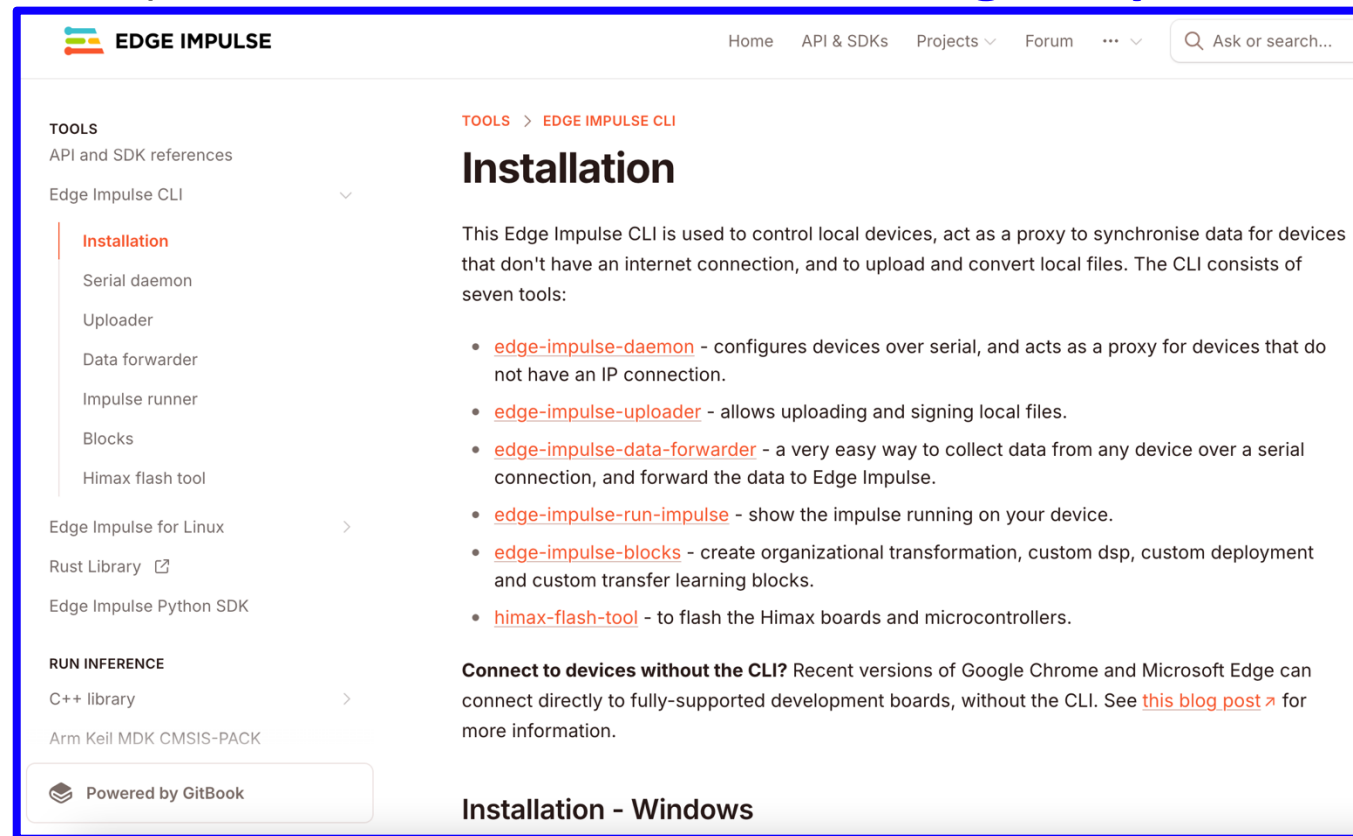
Lab 4 [Hardware] – Setting up Arduino CLI

- Go to <https://docs.arduino.cc/arduino-cli/installation/>
- **Carefully follow instructions** corresponding to your Laptop's/PC's operating system (Windows, MAC, Linux) for **Download and Install Arduino CLI**



Lab 4 [Hardware] – Setting up Edge Impulse CLI

- Go to <https://docs.edgeimpulse.com/docs/tools/edge-impulse-cli/cli-installation>
- **Carefully follow instructions** corresponding to your Laptop's/PC's **operating system** (Windows, MAC, Linux) for **Download and Install Edge Impulse CLI**



The screenshot shows the Edge Impulse CLI installation page. The left sidebar contains a 'TOOLS' section with a dropdown menu for 'Edge Impulse CLI' which is expanded to show 'Installation', 'Serial daemon', 'Uploader', 'Data forwarder', 'Impulse runner', 'Blocks', and 'Himax flash tool'. Below this are links for 'Edge Impulse for Linux', 'Rust Library', and 'Edge Impulse Python SDK'. The 'RUN INFERENCE' section includes 'C++ library' and 'Arm Keil MDK CMSIS-PACK'. The main content area is titled 'Installation' and describes the CLI's purpose. It lists seven tools: `edge-impulse-daemon`, `edge-impulse-uploader`, `edge-impulse-data-forwarder`, `edge-impulse-run-impulse`, `edge-impulse-blocks`, and `himax-flash-tool`. A note mentions connecting to devices without the CLI using Google Chrome or Microsoft Edge. The page is powered by GitBook.

EDGE IMPULSE Home API & SDKs Projects Forum ... Ask or search...

TOOLS
API and SDK references
Edge Impulse CLI
Installation
Serial daemon
Uploader
Data forwarder
Impulse runner
Blocks
Himax flash tool
Edge Impulse for Linux
Rust Library
Edge Impulse Python SDK

RUN INFERENCE
C++ library
Arm Keil MDK CMSIS-PACK

Powered by GitBook

Installation

This Edge Impulse CLI is used to control local devices, act as a proxy to synchronise data for devices that don't have an internet connection, and to upload and convert local files. The CLI consists of seven tools:

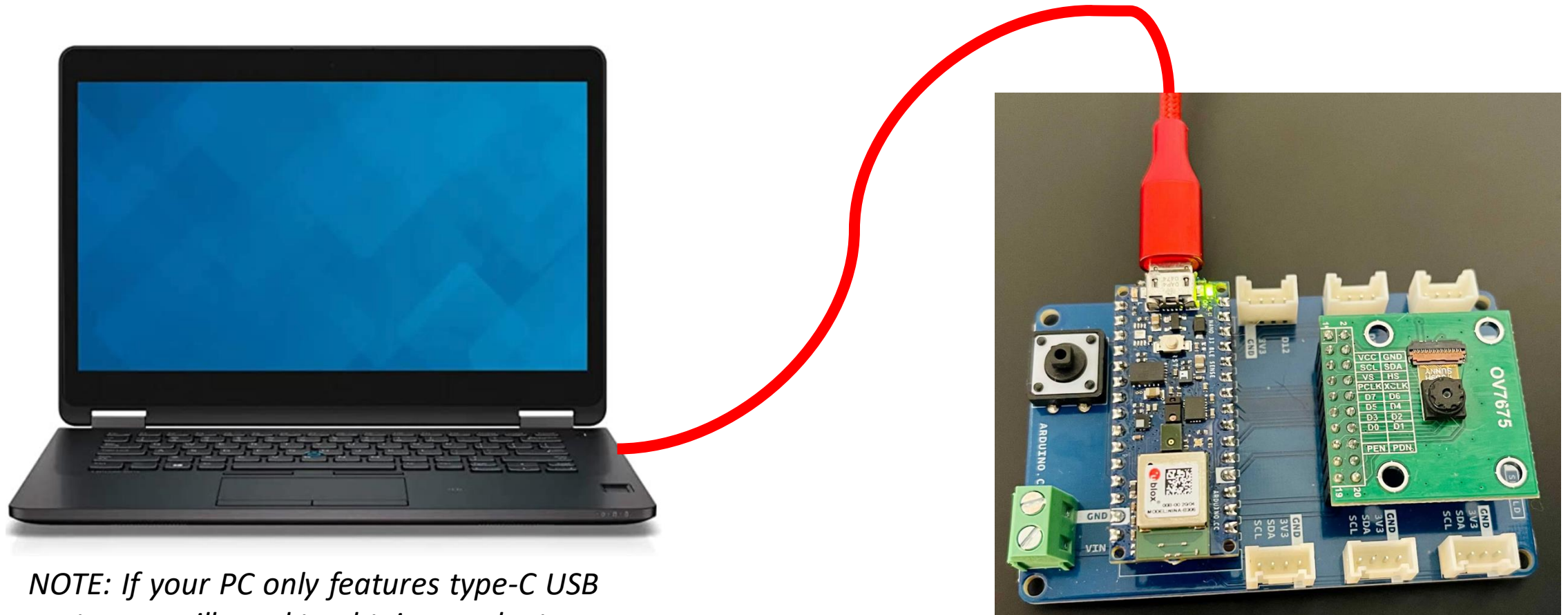
- `edge-impulse-daemon` - configures devices over serial, and acts as a proxy for devices that do not have an IP connection.
- `edge-impulse-uploader` - allows uploading and signing local files.
- `edge-impulse-data-forwarder` - a very easy way to collect data from any device over a serial connection, and forward the data to Edge Impulse.
- `edge-impulse-run-impulse` - show the impulse running on your device.
- `edge-impulse-blocks` - create organizational transformation, custom dsp, custom deployment and custom transfer learning blocks.
- `himax-flash-tool` - to flash the Himax boards and microcontrollers.

Connect to devices without the CLI? Recent versions of Google Chrome and Microsoft Edge can connect directly to fully-supported development boards, without the CLI. See [this blog post](#) for more information.

Installation - Windows

Lab 4 [Hardware] – Connect your TinyML kit

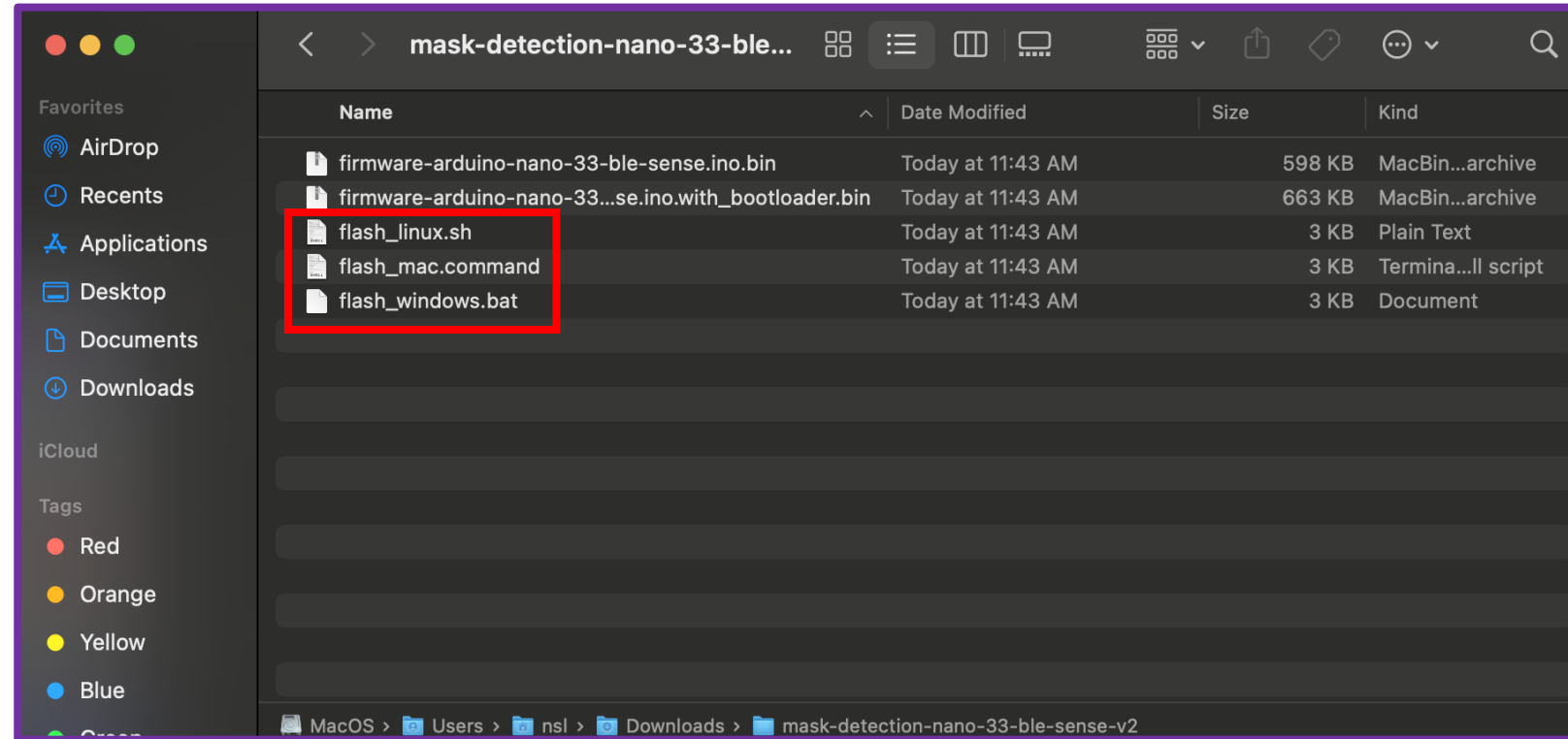
- Use the USB cable (type-A to microB) to connect the Nano 33 BLE Sense to your machine



NOTE: If your PC only features type-C USB ports, you will need to obtain an adaptor

Lab 4 [Hardware] – Flash Your Model to TinyML Kit

- Flash the compressed model (**downloaded from Edge Impulse**) onto the board by **double clicking on the script for your operating system**



Lab 4 [Hardware] – Flash Your Model to TinyML Kit

- **NOTE:** You may need to type the prompted command **“edge-impulse-run-impulse”** in a new command window

```
nsf — flash_mac.command — 80x24
Planes      : 1
Lock Regions : 0
Locked       : none
Security     : false
Erase flash

Done in 0.001 seconds
Write 591376 bytes to flash (145 pages)
[=====] 100% (145/145 pages)
Done in 23.178 seconds
New upload port: /dev/cu.usbmodem143301 (serial)

Flashed your Arduino Nano 33 BLE development board.
To set up your development with Edge Impulse, run 'edge-impulse-daemon'
To run your impulse on your development board, run 'edge-impulse-run-impulse'
logout

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

Lab 4 [Hardware] – Flash Your Model to TinyML Kit

- Follow the instructions prompted in command window to observe the results in the command line

```
ns1 — -bash — 80x24
ns1 — flash_mac.command — 80x24

Planes      : 1
Lock Regions : 0
Locked       : none
Security     : false
Erase flash

Done in 0.001 seconds
Write 591376 bytes to flash (145 pages)
[=====] 100% (145/145 pages)
Done in 23.178 seconds
New upload port: /dev/cu.usbmodem143301 (serial)

Flashed your Arduino Nano 33 BLE development board.
To set up your development with Edge Impulse, run 'edge-impulse-daemon'
To run your impulse on your development board, run 'edge-impulse-run-impulse'
logout

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

